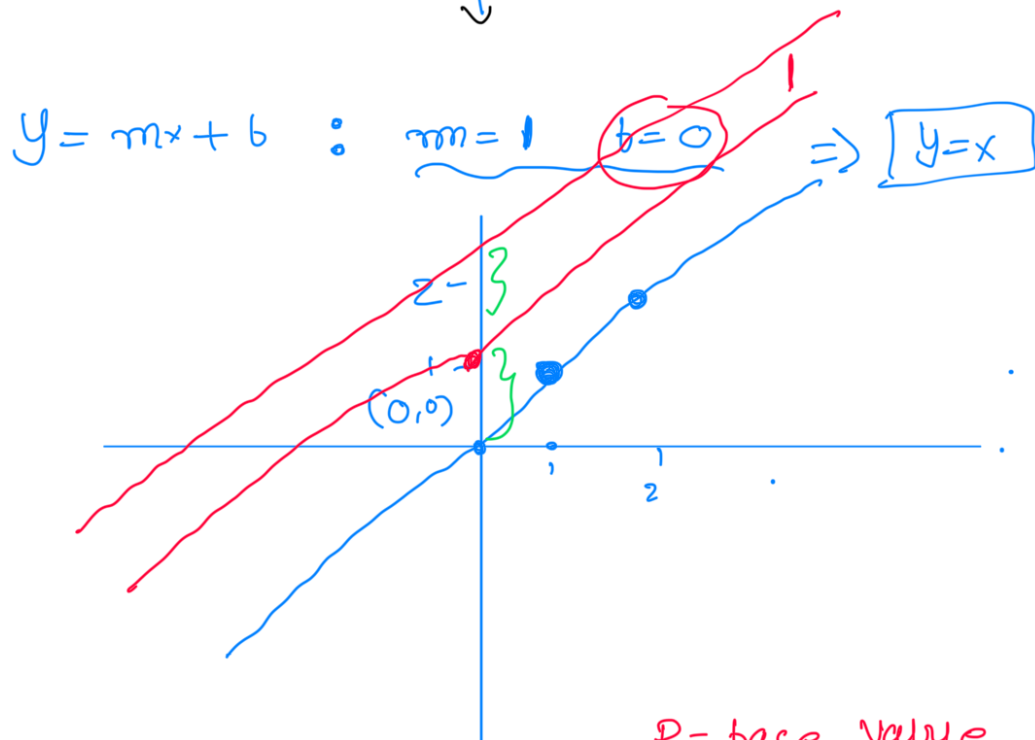
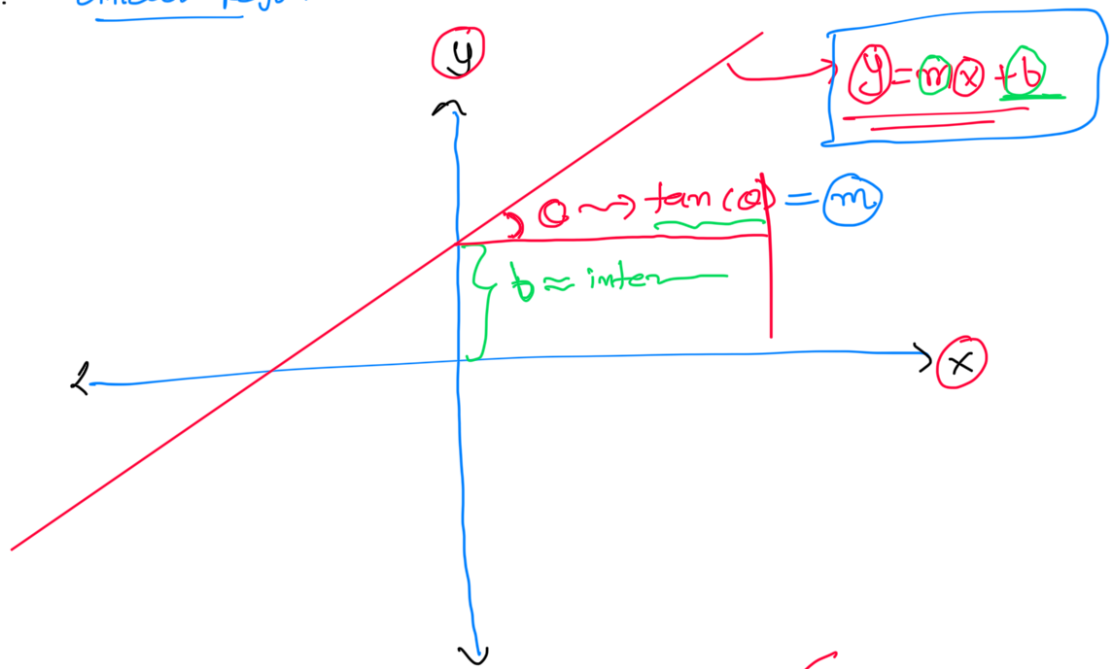


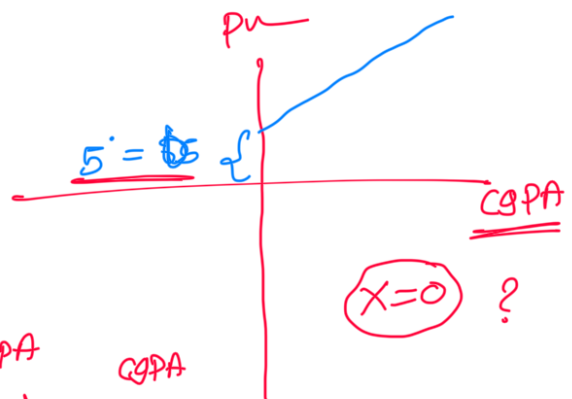
Geometry: Linear Regs.



$y = x$

x	y
0	0
1	1
2	2

B = base value



PKL

LPA
↓
CGPA
↓
 $y = mx + b$

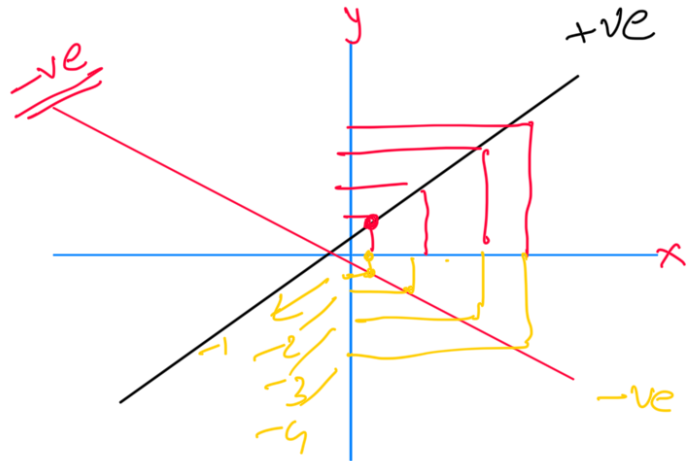
1.

↓ ↗ (5)
 $\boxed{COPA(x) = 0}$

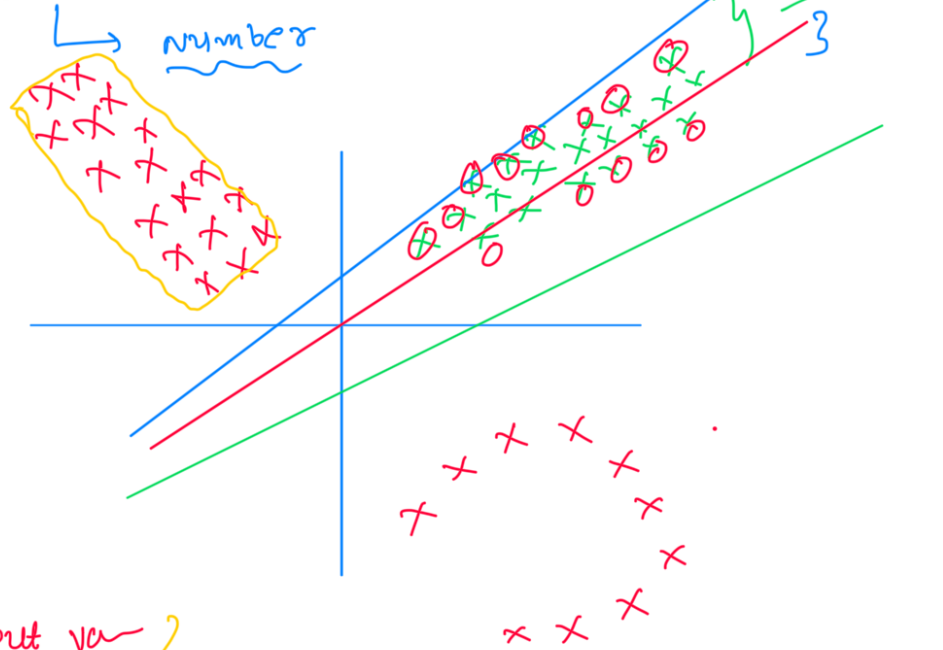
$\boxed{y = 5}$

slope = +ve
slope = -ve

-4 & -1



* Simple Linear Per:



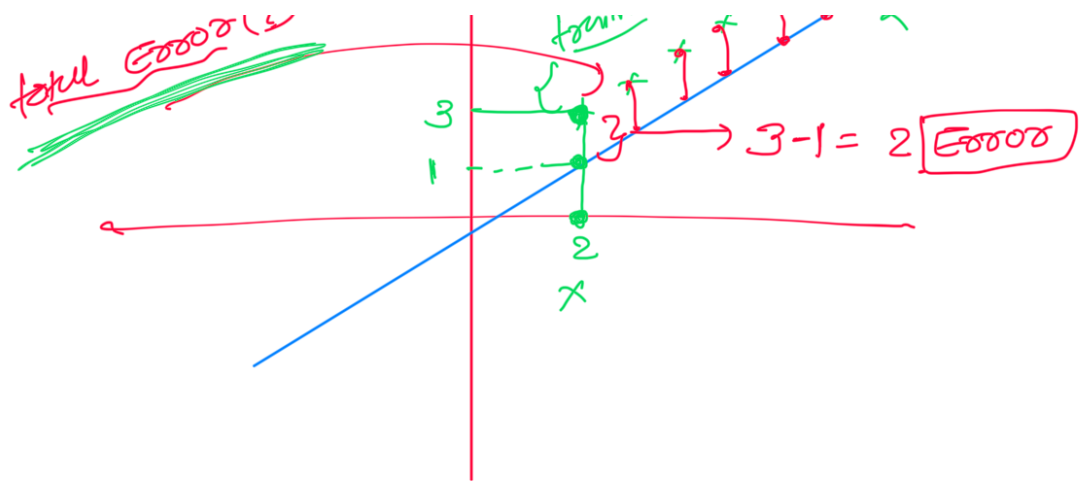
Simple : Input var
Per

COPA } in h

↳ CPA } target

(9)

... in ... } $y = mx + b$



8:

$$y = mx + b \quad \begin{matrix} \text{COPA} \\ \text{LPA} \end{matrix}$$

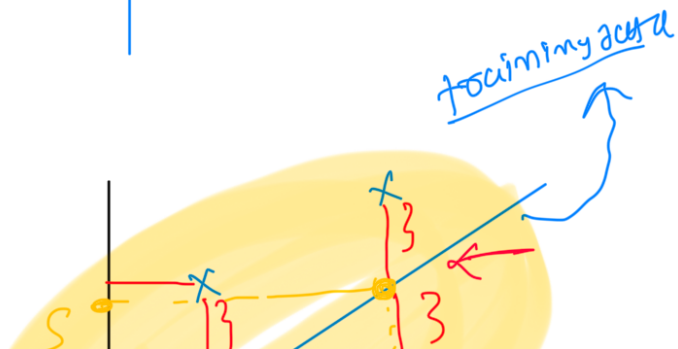
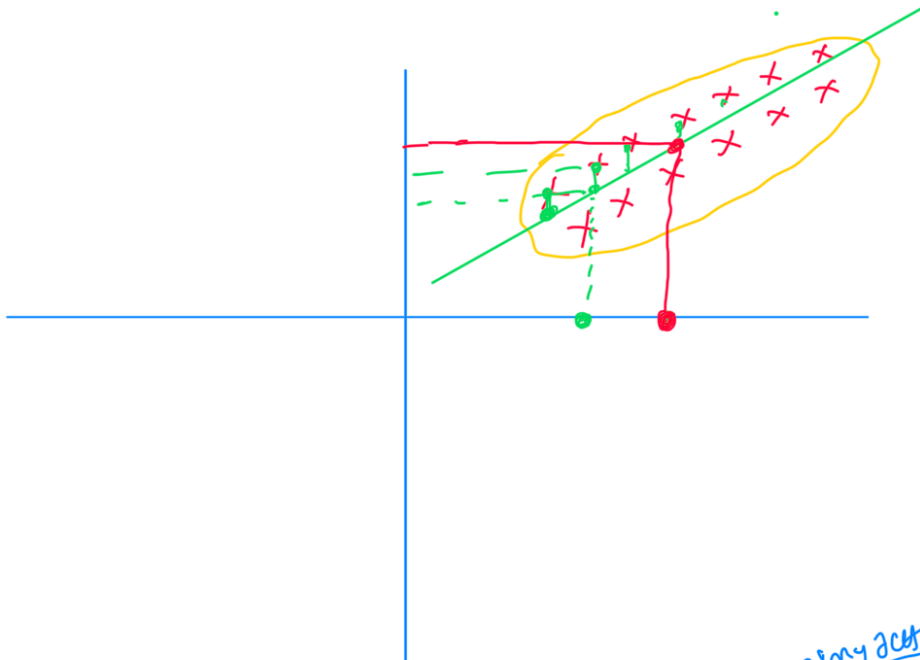
$$\begin{matrix} m = 5 \\ \hline b = 2 \end{matrix}$$

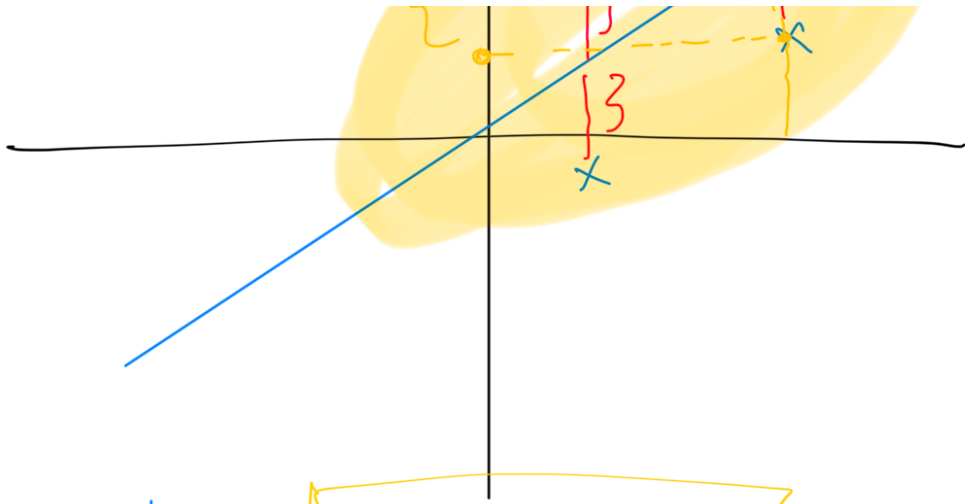
$$\begin{matrix} \text{COPA: } 5 = x \\ \text{LPA: } (?) \end{matrix}$$

$$= 27$$

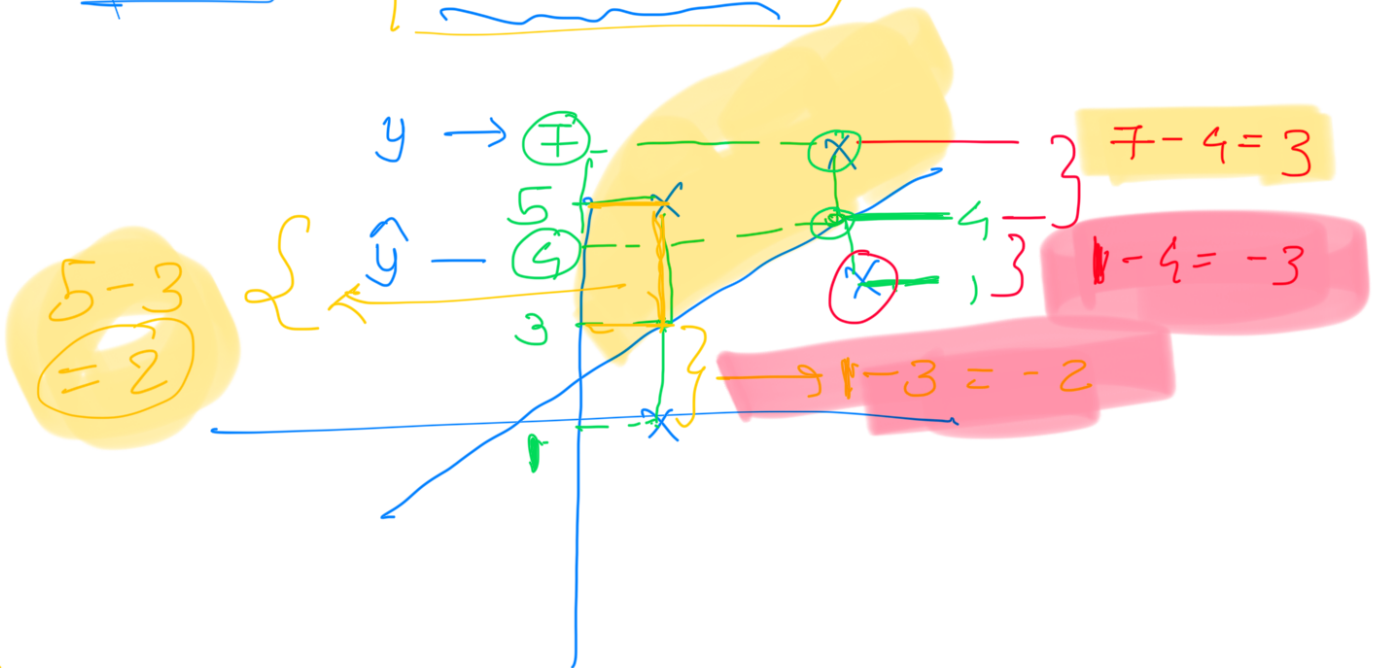
$$y = 5x + 2$$

$$= 25 + 2 = 27$$





Gradient : Error Sum



$$|3| + |-3| + |2| + |-2| = 0$$

→ 10 ✓
① Abs Error

$$(+ve)^2$$

$$(-ve)^2$$

(+ve)

$$8 + 8 + 4 + 4 = \boxed{26}$$

Squared Error

Distance to Delhi: 30 km

$$30 \times 1000$$

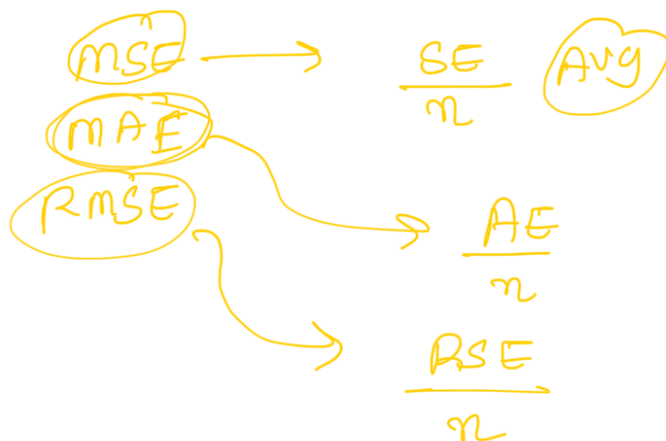
$$30000 = \text{cm}$$

cm

→ MSE

→ MAE

$$\text{RMSE} = \sqrt{\text{SE}}$$



$\hat{y}$ = Predicted var

y = og varu.

$$\text{total} = \sum (\hat{y} - y) \rightsquigarrow \text{zero}$$

$$\text{MAE} = \sum |\hat{y} - y| \quad \text{?} \quad \text{MSE} = \frac{\sum (\hat{y} - y)^2}{n}$$

$$TSE = \sum (\hat{y} - y)^2$$

$$TRSE = \sqrt{\sum (\hat{y} - y)^2}$$

$$MAE = \frac{\sum |\hat{y} - y|}{n}$$

$$RMSE = \sqrt{\sum (\hat{y} - y)^2}$$

total data points $\rightarrow n$

R^2

data $\rightarrow$ train & test

$$y = mx + b$$

$$m = 4$$

$$b = 2$$

$$y = 4x + 2$$

$$y = 24 + 2 = 26$$

6